

Hi!ckathon

CHALLENGE THE ORDINARY, INNOVATE WITH AI AND DATA

4 Weeks Prep

1 Final Sprint

 Nov 4, 2025 – Nov 25, 2025

 Courses, Career Fair

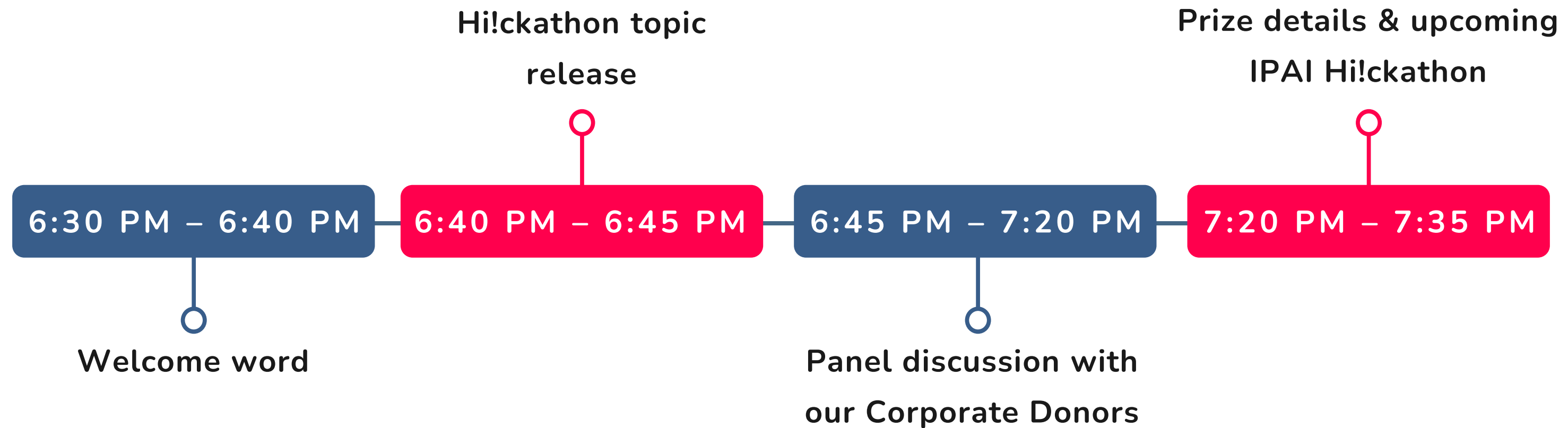
 Nov 28, 2025 – Dec 1, 2025

 Data Challenge, Pitch

Hi!ckathon

OPENING CEREMONY AGENDA

November 28th, 2025





Christophe PERIGNON

Hi! PARIS Scientific Co-Director

Created in September 2020 by **two leading institutions**



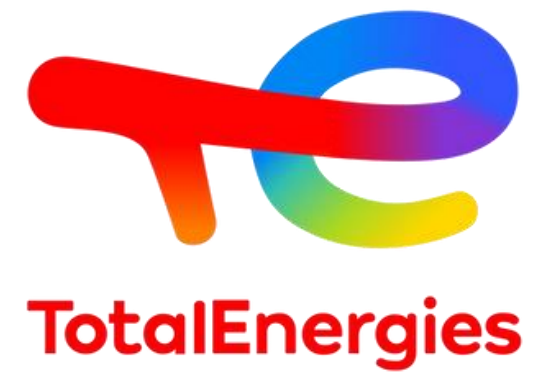
... joined in **2021** by

Inria

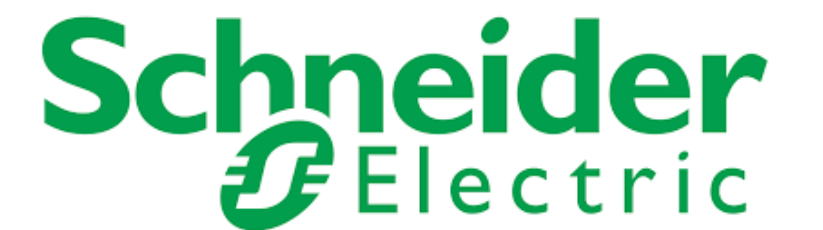
In 2024, **CNRS & UTT** joined Hi! PARIS. The center was officially labeled an **AI Cluster** by the French state, securing **€70M** in **funding**



Backed by leading
Corporate Donors



L'ORÉAL



Research

+200

Faculty members in AI
& Data Science

13

European Research Council

ERC in AI (active since launch)

+430

Articles in top-tier journals
and conferences in AI

Education

+250

PhD students
in AI & Data Science

 **#1**  **#10**

In France Worldwide
Graduate Employability
(QS 2024)

 **#2**
In Europe (FT 2024)

 **#2**  **#41**

In Europe Worldwide
QS World University Rankings 2026

 **#1**  **#2**

In Europe Worldwide
X-HEC MSc Data Science
for Business (QS 2026)

Innovation

50%

of the French
unicorn-founders
are alumni from
our institutions

171

Startups in AI are
founded, incubated, or
accelerated within our
entrepreneurial
ecosystem

Society



Nonprofit

HEC
PARIS



Public status

 INSTITUT
POLYTECHNIQUE
DE PARIS

 ENST2  ENSAE  



Expertise in equal opportunity
actions, particularly targeting
high school pupils

A lookback to the last editions...



Some members of our Team that will be present with you during this challenge



Pierre-Antoine Amiand Leroy
ML Research Engineer



Nassima Ould Ouali
ML Research Engineer



Awaid Hussain Sani
Senior ML Research Engineer

+10

Capgemini's
Support Team



Soobash Daiboo
Senior Data Scientist



Thibault Porssut
Senior Data Scientist



Nathan Neike
Junior Data Scientist



Damien Ngo
Junior Data Scientist

& many more colleagues are also here, ready to assist you whenever needed!

Hi!ckathon

CHALLENGE THE ORDINARY, INNOVATE WITH AI AND DATA

Topic announcement



What's the theme?
Can you guess?





Vincent FRAITOT

Professor - MSc Data Science and AI for Business HEC | X Director

AI & Education

From PISA data to an
innovative AI solution

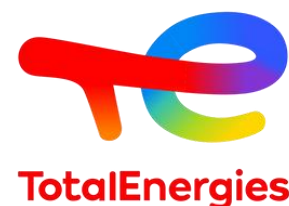
Opening Ceremony Roundtable with Our Corporate Donors

L'ORÉAL



Stéphane Lannuzel

SVP Technology,
Data, AI



Xavier Morise

Scientific Advisor for
R&D and International
Partnerships



Artus Mathieu

Data Scientist
@Leonard



Ania Kaci

Global Leader for
Responsible AI



Vincent Fraitot
Moderator

Associate Professor - MSc Data Science
and AI for Business HEC & X Director

TOPIC #1

As AI & data analytics transform industries, how would you describe the comprehensive skill set and personality traits of an exemplary Data “profile” suited for current business demands?

TOPIC #2

How can data analytics and AI contribute to making education more personalized, inclusive, and effective both in schools and within corporate training programs?

TOPIC #3

What would be your expectations for the Hi!ckathon and advice for the students?

About PISA

The programme goes beyond evaluating purely theoretical knowledge : it examines how young people are able to use what they have learned in real-life situations. It also considers external factors such as the socioeconomic and cultural background of families, the school environment, and the national education system, as well as internal factors like students' motivation, self-esteem, and learning strategies. The ultimate goal is to identify the key factors of success in order to help countries improve their education systems.

Dataset

The dataset provided for the hackathon merges the 2015, 2018, and 2022 PISA cycles into a single unified table containing roughly 1.7 million student records and more than 250 variables. It includes student-questionnaire features that have been cleaned and transformed from raw survey responses into ready-to-use numerical data, then organised into thematic groups to facilitate analysis. The dataset also integrates cognitive test information, including item-level scores and response times.

1st Task, a Regression Task

Build models that predict each student's **MathScore**. The goal is to explore how background factors, learning habits, school climate, and item-level behaviour relate to math performance. Teams are encouraged to focus on interpretability and insights that can support educational decision-making.

2nd Task : Innovation

The dataset is very detailed, containing a wealth of information on a wide variety of topics. Your role is to build a data-driven solution to address a specific issue. Using the dataset, you must justify your approach with graphs, build a business plan, design an innovative solution, and present it with a pitch.

You must choose one issue from the four possible issues (see following slides).

Objectives and Deliverables

Scientific & Business Approach

You need to justify your predictions, explain your solution, demonstrating how your approach is innovative, viable and socially profitable.

Presentation “Pitch” goal

In 3 minutes, you must explain your project, demonstrating its viability, showing how your proposal is innovative and original, and what the added value to society is.



1

Strengthening academic engagement: how can we improve students' concentration and motivation?

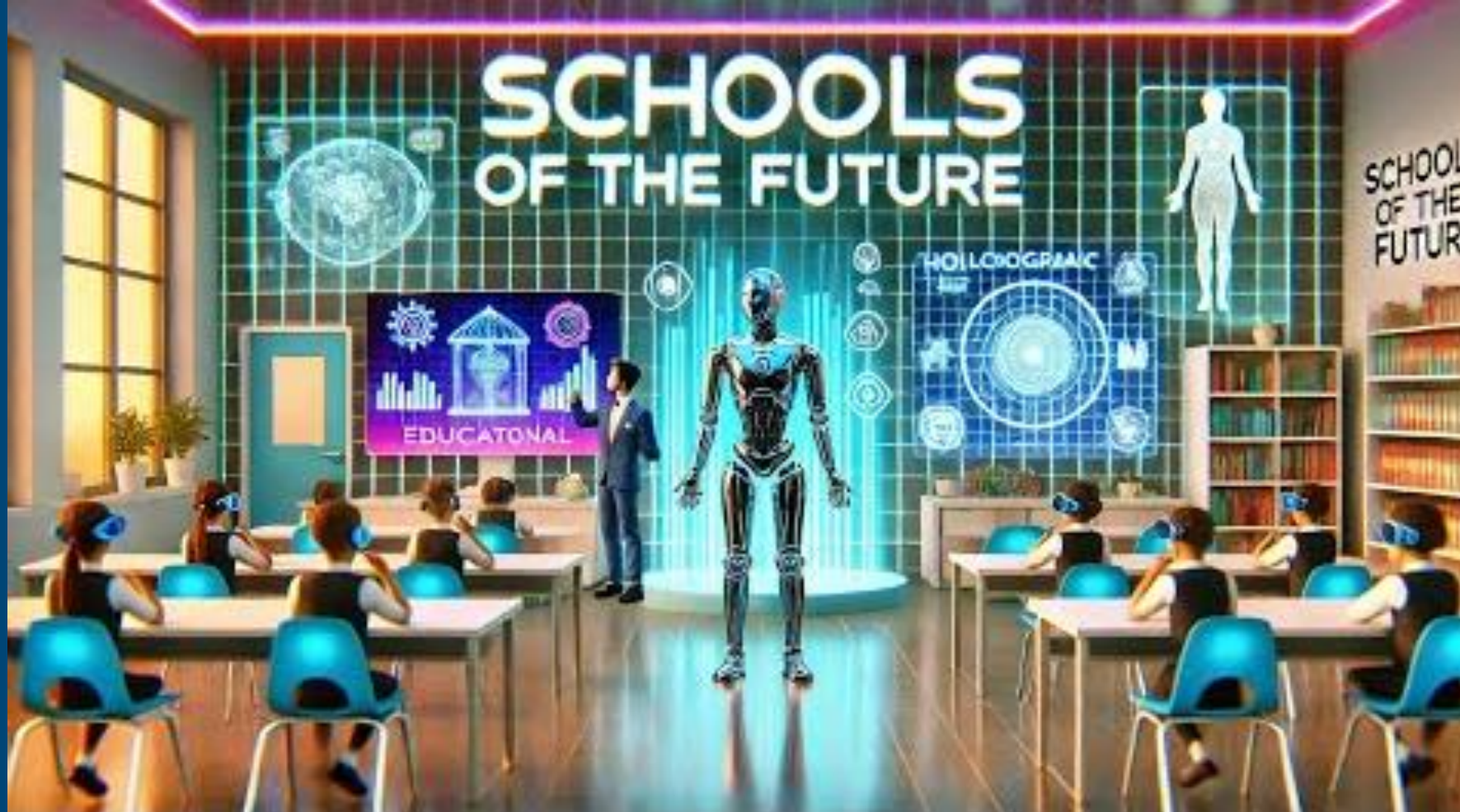
Concentration and motivation play a key role in academic success. Many factors (digital distractions, personal difficulties, lack of confidence, attention disorders, family environment) can impact a student's ability to stay engaged. In a context where teaching methods are evolving and average attention spans are tending to decrease, it is becoming essential to offer solutions that stimulate students' interest and involvement from an early age.



2

Ensuring educational continuity: how can we support students when their relationship with teachers has been weakened?

Students are more prone to school failure when the interaction between teachers and learners is weakened or broken (due to geographical distance, illness, health crises, absenteeism, or staff shortages). To ensure that knowledge is properly transmitted, how can we maintain or rebuild strong teacher–student connections under such conditions?



3

School of the future: How can we ensure equitable mastery of digital skills in the age of AI?

Education is changing fast: digital tech is everywhere, online resources are booming, AI tools are popping up all over the place, and new teaching methods are being tried out. However, digital tools can either support or impede learning depending on how they are used by learners and by teachers. The digital skills gap is widening with every new technology. Moreover, not all students and teachers have the same access to this technology: quality of internet connection, accessibility to good computers, overuse of smartphones... These disparities increase inequalities among teachers and students. Solutions are to be found to ensure a fair transition.



4

Mental health & well-being: how can we come up with innovative solutions to support students?

Mental health issues are increasingly prevalent in schools: stress, anxiety, cyberbullying, social isolation, academic pressure, etc. A student experiencing distress may see a decline in their motivation, performance, and self-esteem. New approaches are emerging (prevention, digital support, self-assessment tools, discussion forums), but there is still much to be done to effectively support young people.

Five Thematic Awards!



Scientific approach

Rewards the team which best demonstrates the ability to **justify its scientific & business approach** and to confront the **state of the art** in data science



Technical excellence

Rewards the model with the **highest score**. The score is a synthetic measure of the model's accuracy and code quality.



Interdisciplinarity

Rewards the teams able to combine several disciplines in the production of the **scientific & business approach and presentation**.



Business opportunity

Rewards the team that best projected itself in a **market / real-world context** through an **impactful pitch**.



Public's prize

Rewards the **public's favorite team** during the live pitch session.

EXCLUSIVE

70

Students

2

Countries

35

from France

35

from Germany

#1

Mission

Connecting Ideas. Crossing Borders. Creating Impact.

EXCLUSIVE

HI!CKATHON INTERNATIONAL EDITION: FRANCE–GERMANY COLLABORATION

This year, we're joining forces with the **IPAI Foundation** to launch the **first International Edition of the Hi!ckathon**, an initiative that connects students, ideas, and perspectives across borders.

Starting in Paris and continuing in **Heilbronn, Germany**, in March 2026, 70 students: **35 from France and 35 from Germany**, will explore this question from two complementary perspectives, combining Data Science, innovation, and societal impact.



The **35** students representing France will be selected during this **Hi!ckathon!**

Timeline of the event



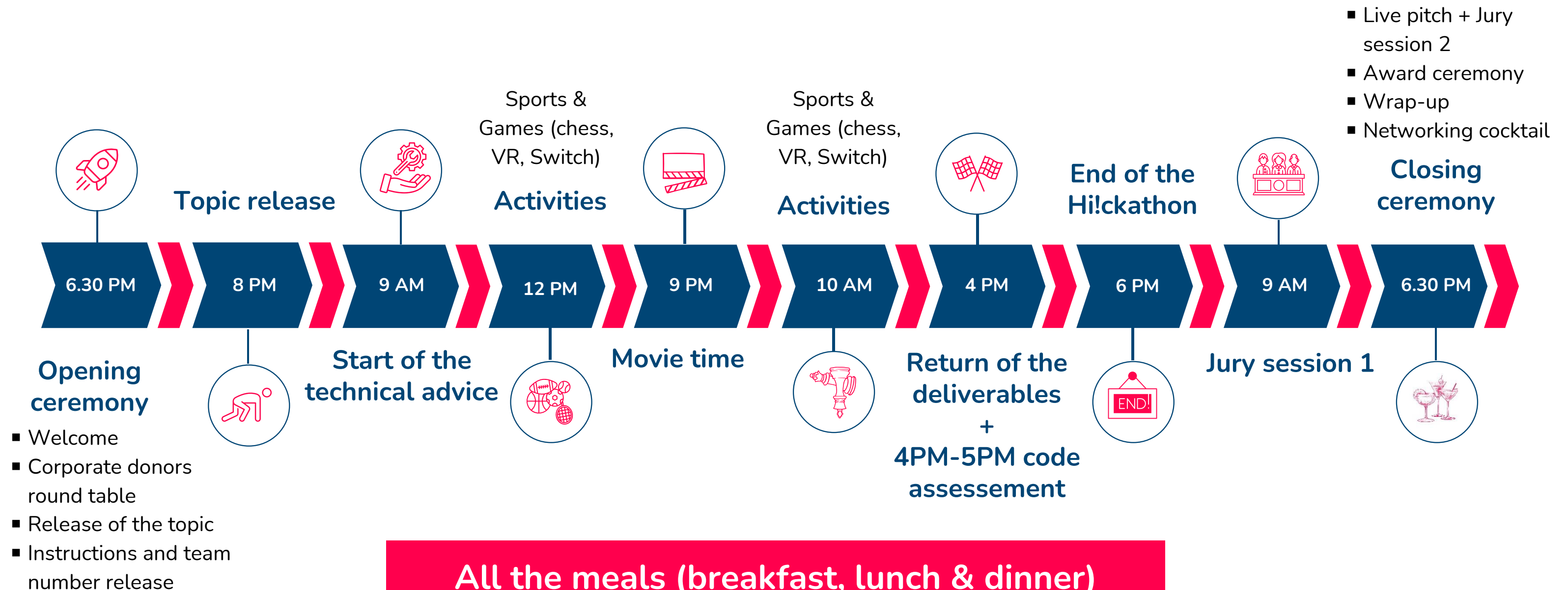
All activities will be held on site

Friday 28th
ENSTA

Saturday 29th
ENSTA

Sunday 30th
ENSTA

Monday 1st
ENSTA



**All the meals (breakfast, lunch & dinner)
are covered during the weekend**

Tools to use

Access to the challenge topic
and datasets



Communicate



Code your model



Any question?

Follow the blue shirts/hoodies and contact us on Discord!



Step-by-step guide on the hfactory platform

- 1 Register to the Hi!ckathon #6 activity on HFactory (you should have received an invitation by email)



Your platform for engaging learning experiences

Activity-based learning in Data Science and AI.

Get started now >



An email with the code has been sent to alioune.ka@capgemini.com.

425062

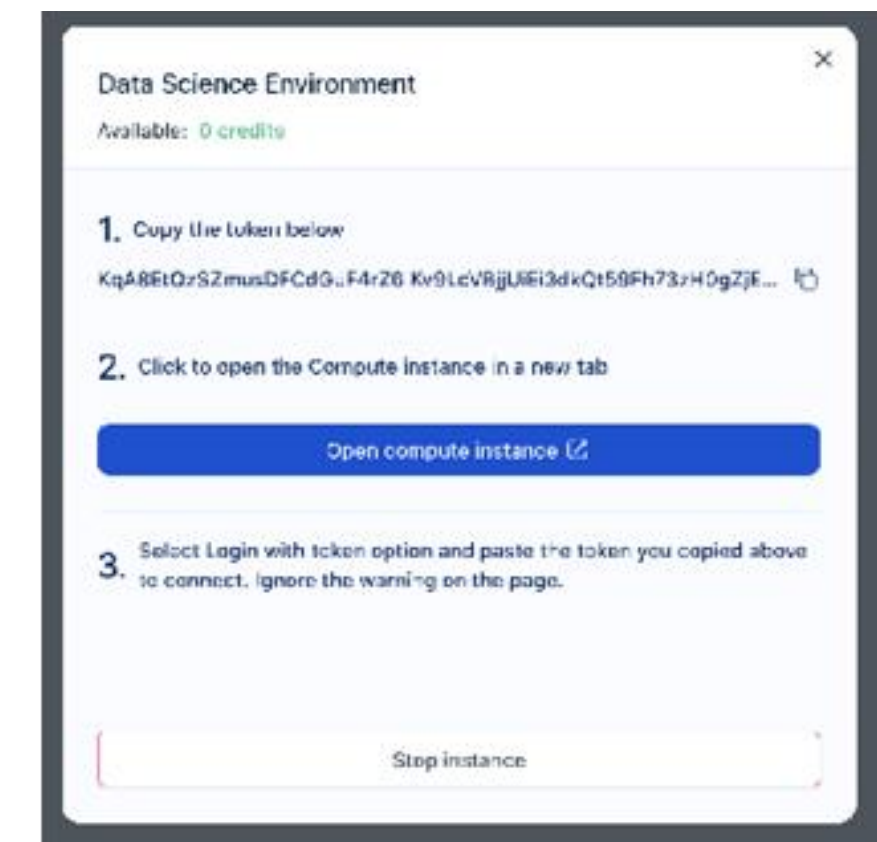
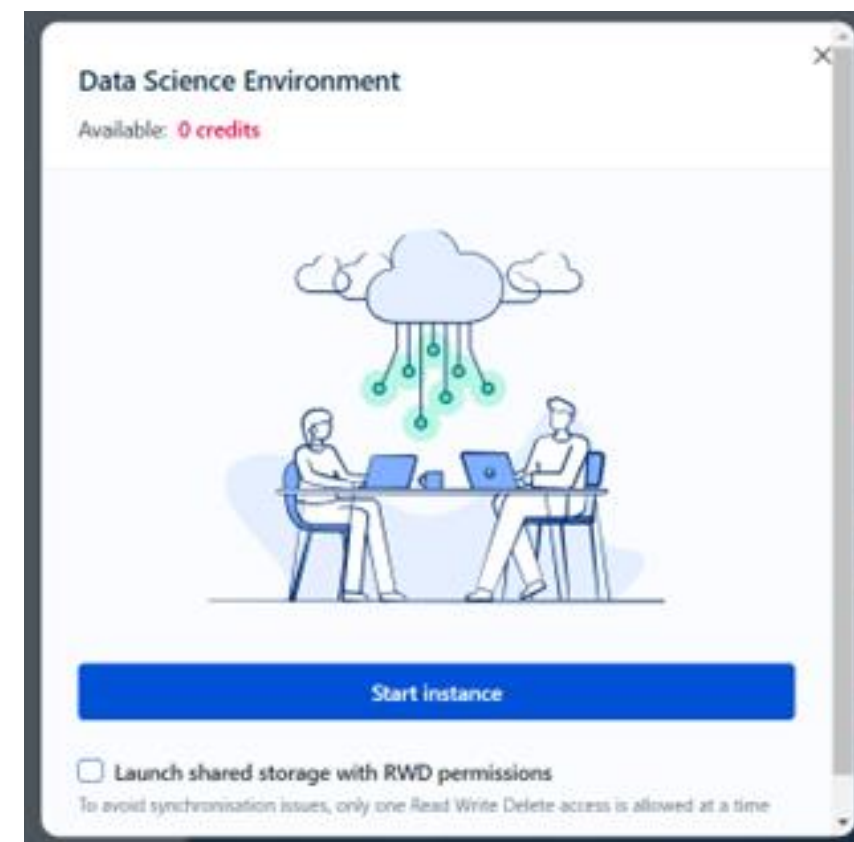
Did not get the code?

Submit >

- 2 Click on the compute icon in the top menu bar to access the Data Science computing environment

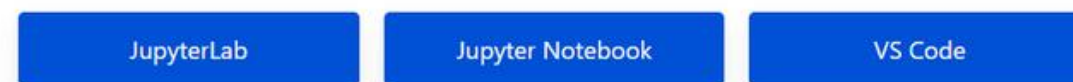


- 3 Start a CPU/GPU computing instance (note that GPU resources are limited) Then, copy the token then click on "Open compute instance" to login with it.

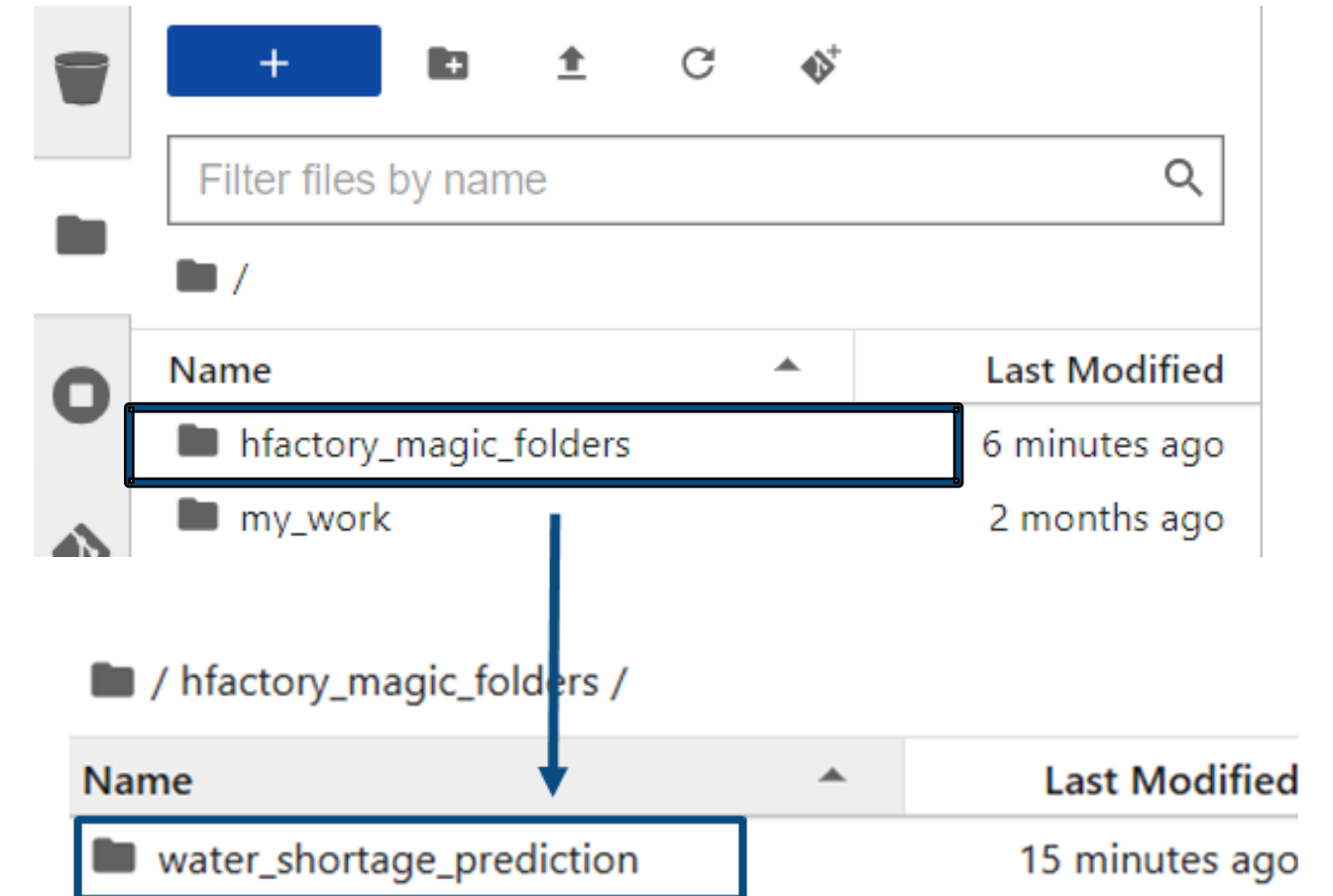


Step-by-step guide on the hfactory platform

- 4 Now, you can select an IDE to use with the computing environment (JupyterLab, Jupyter Notebook, VS Code)



- 5 Once you've selected an IDE, you can open the computing environment. In this environment, you can access the dataset and work in your own or group environment.



Let's stay in touch



/hi-paris-center



Any question?

Are you ready?



8





Hi!ckathon #6

Student guide



What is the Hi!ckathon ?

The event

Competitive Data / AI event of Hi! PARIS, the interdisciplinary research center for Artificial Intelligence founded by Institut Polytechnique de Paris and HEC Paris.

The theme

AI & Education

Beyond producing an AI model, the competition will ask you to realistically project your solution in a market / real-world context.

When?

From Friday **November 28th, 6:30pm** to Monday **December 1st, 9pm**.

- Friday to Sunday Evening at **ENSTA Paris**
- Monday Evening (Closing Ceremony) at **ENSTA Paris**.

Timeline of the event



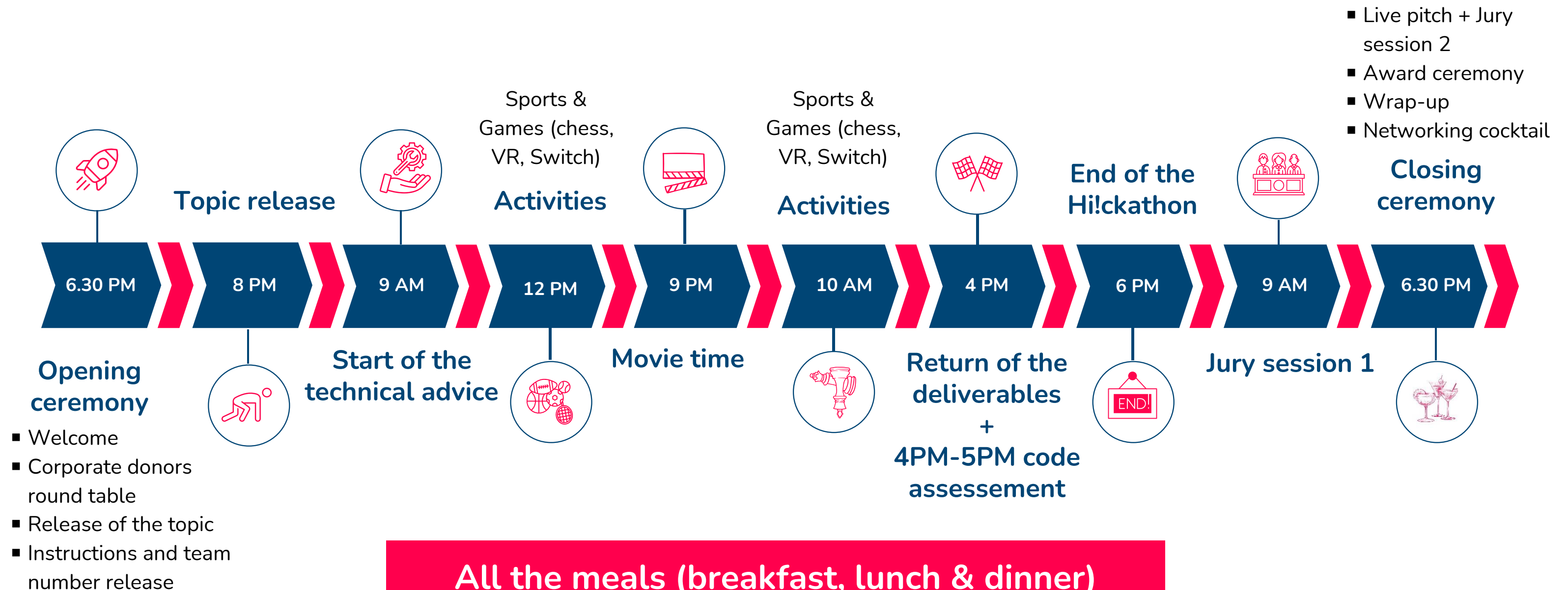
All activities will be held on site

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Saturday 29th
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Sunday 30th
ENSTA

Monday 1st
ENSTA



All the meals (breakfast, lunch & dinner) are covered during the weekend

What are the deliverables ?



Build an AI model and make predictions

Develop a state-of-the-art AI model that addresses the sustainability use case



Create a scientific / business document

Justify the scientific and business approach of your AI solution



Produce a video presentation

Film a short video to present your project and defend its merit



Defend your project in front of a live audience*

Some groups will be asked to present their project in front of a live audience during the closing ceremony.

The groups who will have to present live during the closing ceremony will be notified Sunday evening. More information on this can be found in **Deliverables and deadlines.*

Hi!ckathon awards



Award	Description	Given by
Business opportunity	Rewards the team that best projected itself in a market / real-world context through an impactful pitch .	L'ORÉAL
Scientific approach	Rewards the team which best demonstrates the ability to justify its scientific & business approach and to confront the state of the art in data science	Capgemini
Technical excellence*	Rewards the model with the highest score . The score is a synthetic measure of the model's accuracy and code quality.	TotalEnergies
Interdisciplinarity	Rewards the teams able to combine several disciplines in the production of the scientific & business approach and presentation .	VINCI
Public crush	Rewards the public's favorite team during the live pitch session.	Schneider Electric

*Winners of the **Technical excellence award** will be asked to present their project during the event's closing ceremony. The winning group will be contacted by a member of the organizational team right after the 4PM Sunday submission deadline.

Guidelines for participants



Going to unauthorized floors/areas is prohibited

- Only the ground-floor and first floor (level 0 and 1) are accessible during the competition.
- Areas bared by masking tape cannot be reached by participants.



Please keep your room/designated area clean

- Participants must keep their room/designated area clean during the duration of the competition.
- Teams who don't comply with this rule by the end of the weekend will be penalized or disqualified (the same rule applies for accessing restricted areas).



Need to sleep or take a shower ?

- Sleeping rooms are available to rest during the night (only between 11PM-7:45AM).
- Showers are available at ENSTA Paris during the weekend.



Access to the venue during the weekend

- The main entrance of Télécom Paris will be closed for the whole duration of the final sprint, starting Friday at 8PM.
- To go in or out, you have to pass by the security desk.



Any trouble during the competition ?

- Hi!ckathon #6 team members are available anytime during the weekend, from 8AM to 10PM.
- Don't hesitate to reach out to any of us regarding any issues/concerns.

Guidelines for participants

Where can I eat near the venue ?

- **PANIFIQUE** (TAKEAWAY): 13 Av. de la Vauve, 91120 Palaiseau
Opening hours: from 6:30 am to 8 p.m / Monday to Friday
- **FRANPRIX** (In-store shopping): 13 Av. de la Vauve, 91120 Palaiseau
Opening hours: from 9 am to 9 pm / Monday to Sunday
- **Living Room Palaiseau** (Dine-in / Takeaway): 10 Boulevard Gaspard Monge, 91120 Palaiseau
Opening hours: from 12 pm to 2:30 pm and from 7 pm to 10:30 pm / Monday to Saturday
- **Le Julie's** (Dine-in / Takeaway): 40 Cr. Pierre Vasseur, 91120 Palaiseau
Opening hours: Monday to Friday (7 am to 11:30 pm) , on Saturday (5 pm to 11:30 pm)
- **FOOD ASIA** (Dine-in / Takeaway): 2 Boulevard Gaspard Monge, 91120 Palaiseau
Opening hours: from 11:30 am to 9:30 pm / Monday to Saturday
- **All Together** (Dine-in): 27, boulevard Thomas Gobert - 91120 Palaiseau
Opening hours: from 11:30 am to 2:30 pm then from 6 pm to 10:30 pm / Monday to Friday

For the final sprint 

What resources do I have ?

- A welcome document
- A student guide
- Datasets to train/evaluate your AI solution
- A variable dictionary
- An HFactory guide
- A guide notebook to load the datasets in HFactory

Where can I find these resources ?

- Final Sprint: Go to the « *Hi!ckathon #5 Final Sprint* » activity on HFactory + [Google Drive](#)

Deliverable 1: Data Science model & prediction



What should I submit ?

- Submit the **code** of your Data Science model (in .ipynb or .py format). *Your code will be evaluated by the Hi!ckathon team for code quality.*
- Submit your **predictions** on the test set to the leaderboard (in .csv format). *Note that you can only make up to ten submissions to the leaderboard.*

How is code quality evaluated ?

- **Executability:** You will be asked to run your code live in front of Data Science coaches during code assessment. To speed-up the assessment, we will only ask you to run the code on 10 000 rows. Groups that don't show up or can't run their code live will be **disqualified**.
- **Code structure:** Good use of functions/classes to structure your code. Adding comments, docstrings and/or markdown cells to document your code.

Deliverable 2: Scientific / Business approach



Requirements / Target

- Submit a short PDF document detailing the **scientific** and **business** approach you developed for your project/solution. 6 pages maximum.
- Make your document **clear and concise**. It shouldn't exceed a 4 minutes reading time (maximum 4 pages). Adding more pages will lead to the disqualification of your group.
- Don't hesitate to add **visualizations** to make your approach engaging and easily understandable to the jury.

Deliverable 2: Scientific / Business approach



Mandatory sections

I. Overview

- Problem statement (What is the issue ? Why is it critical ?)
- Presentation of your project / solution
- Presentation of your team

II. Business approach

- What is the value of the solution ?
- How does it integrate into real-world operations?
- How is it different from other solutions ? What makes it novel ?

III. Scientific approach

- What is the technical foundation of your solution ? (pre-processing, model)
- How did you make your approach environmentally frugal ? (can also be answered in the business approach)
- How does it scale ?

IV. Results and Future Potential

- What results have you achieved ? (score, explainability)
- What is the future potential of your solution ?

Deliverable 3: Presentation video



Requirements / Target

- Create a **short, facecam video** presenting your solution.
- The video must **not exceed 3 minutes**. Videos longer than 3 minutes will result in disqualification.
- **No editing or cuts** are allowed, but you can use props if you want to.
- This video will be used to select the qualified groups for the final live pitch session held during the closing ceremony.
- **You can use only 1 slide without animation**



Deliverable 3: Presentation video

Here are some excerpts of the business pitch course :

Sharing your Impact-Making Idea

A good presentation is not an option:

- 7% words
- 38% tone of voice
- 55% body language



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Explain your Solution

- Max 14-15 words
- Simple & understandable way (visualise/prototype)
- Provide an explanation but not technical details
- If needed you can use more than one slide

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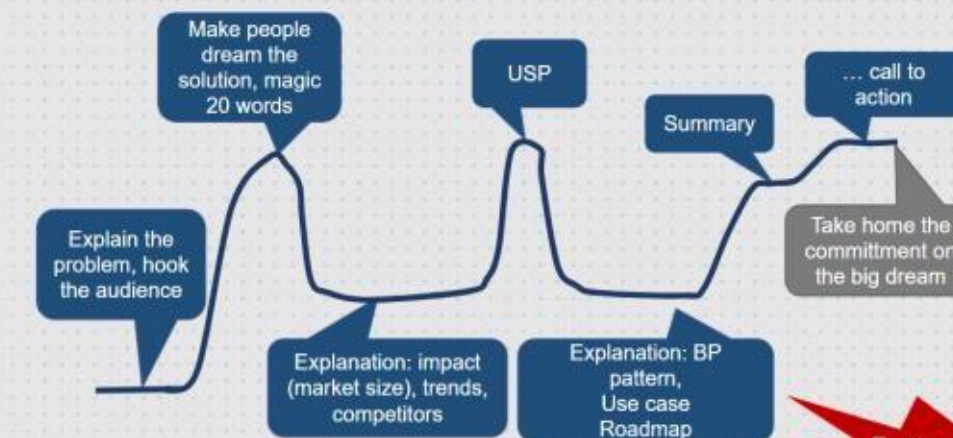
Why You???

- What is special in your solution? Why shall we choose you and not the others?
- What do you do better than your competitors (USP)

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Resonate

Manage the tone of your presentation ... like a storyteller



Remember, 2-3 minutes!

Look at http://www.ted.com/talks/gill_bottle_taylor_s_powerful_stroke_of_insight.html
Look also at http://www.ted.com/talks/nancy_duarte_the_secret_structure_of_great_talks.html
Advised reading: «Resonate» by Nancy Duarte



Deliverable 4: Defend your project in front of a live audience

Requirements / Target

- Present your project to a live audience/jury during the Hi!ckathon's closing ceremony
- Same as the video, **your speech shouldn't exceed 3 minutes**.
- You can deliver the same presentation as the video or modify it if you want to.
- **This deliverable only concerns 10-14 groups** who will be notified Sunday evening of their participation.
- The live presentation is a **component of the jury's evaluation to win awards**. *However, not being selected to perform the live pitch does not disqualify you from winning an award.*
- Please prepare a **single slide** to present your project for the live presentation.



Deadlines and disqualification

Important deadlines

- The **code**, **scientific/business approach** and **presentation video** should be submitted before **Sunday at 4PM** via the *submission form* on HFactory
- Don't forget to attend the **code assessment** from **4PM to 5PM** on Sunday. You will be asked to run your code live (data preprocessing + inference)

Disqualification

- Groups must **submit all required deliverables** and adhere to **all specified requirements** to avoid disqualification. This includes mandatory participation and/or attendance in both the **code assessment** and **live presentation** sessions.
- Any form of **plagiarism** will also to disqualification (for all parties involved). *For students whose participation is mandatory, your pedagogic director will also be notified of any plagiarism.*

Where do I make submissions ?

All of the submissions should be made on the “Hickathon #6 – Final Sprint” **HFactory activity**.

- Submit your **test predictions** to the **Leaderboard** section
- Submit all remaining files (**python code, scientific/business paper, video pitch**) to the **Submissions** section

The screenshot shows the HFactory submission interface. On the left, there is a sidebar with three items: 'About the use case', 'Leaderboard', and 'Submissions'. To the left of this sidebar, there are two dark blue boxes with white text: 'Predictions' and 'Code, paper, video'. Arrows point from these boxes to the 'Leaderboard' and 'Submissions' items in the sidebar, respectively. The main content area is titled 'Submission' and includes a 'Group submission' button. It displays the deadline for the assignment as 'Sun. Dec 1st, 2024 at 4:00pm' and the time remaining as '2 day(s) / 4 hour(s) / 19 min(s)'. Below this, a section titled 'What should you submit ?' lists three items: 'The Python code (in .ipynb or .py format) you used to train your model and make predictions to the leaderboard', 'The Scientific/Business Paper in PDF format (maximum 4 pages)', and 'The video business pitch (maximum 3 minutes)'. A note states: 'Your model predictions should be submitted to the leaderboard page, not this one.' Below this, a section titled 'Guidelines for submissions:' provides further instructions: 'You only need to submit one of each deliverable per group (submissions are at a group level). Please include your group number in the file names you submit (for example: video_pitch_hi5_group0.mp4)'. At the bottom, there is a section titled 'Add your submission files here:' with a dashed box containing the text 'Add a new file', an upload icon, and the text 'Drag your file here or click to upload'.

Predictions

Code, paper, video

About the use case

Leaderboard

Submissions

Submission

Group submission

Deadline for the assignment: Sun. Dec 1st, 2024 at 4:00pm

Time remaining: 2 day(s) / 4 hour(s) / 19 min(s)

What should you submit ?

- The Python code (in .ipynb or .py format) you used to train your model and make predictions to the leaderboard
- The Scientific/Business Paper in PDF format (maximum 4 pages)
- The video business pitch (maximum 3 minutes)

Your model predictions should be submitted to the leaderboard page, not this one.

Guidelines for submissions:

You only need to submit one of each deliverable per group (submissions are at a group level). Please include your group number in the file names you submit (for example: video_pitch_hi5_group0.mp4)

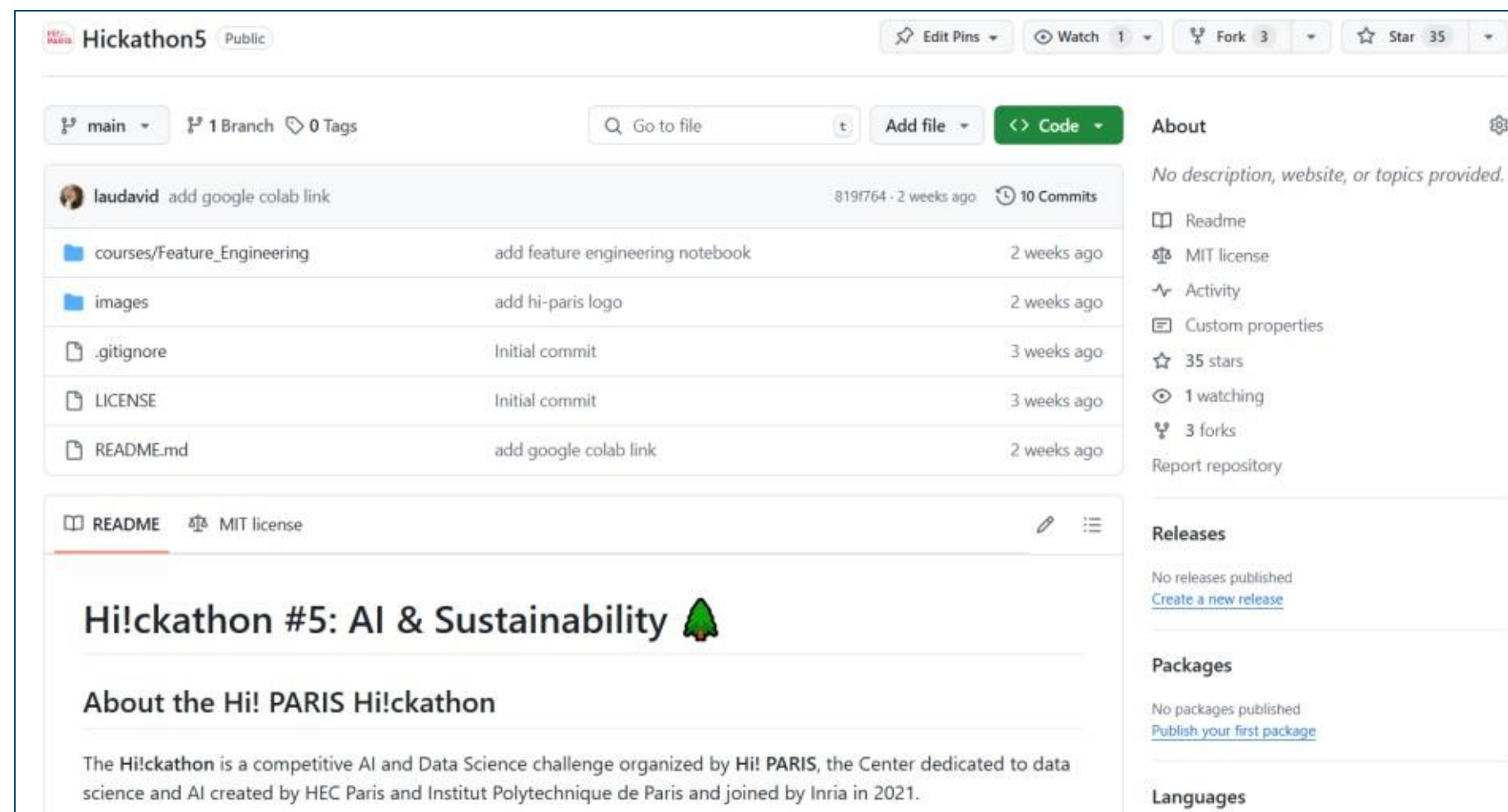
Add your submission files here:

Add a new file Drag your file here or click to upload

Share your project on github

The Hi! PARIS center adopts an **open-source distribution logic**, according to which your AI solution can be shared to the Hi!ckathon #6's public github repository.

If you want to share your work on a personal github, we ask that you mention in your README.md the Hi!ckathon #6 github repository.



Coaching during the weekend

Business coaching

- Q&A sessions with business coaches from Hi! PARIS's corporate donors will be organized **Saturday** and **Sunday**.
- Each team will be assigned to a Q&A session at a **specific time slot**. Not all team members need to join the session.
- Sessions aren't specific to a single team. Multiple teams (but not all) can attend the same session

Technical coaching

- The organizational team is made up of **Hi! PARIS Research Engineers**, as well as **Data Science Consultants**. You can ask us any technical question.
- We are available anytime during the weekend (8AM-10PM), except during sleeping hours.
- We can provide **guidance in understanding the dataset** or in **following a Data Science methodology**. We cannot not intervene directly on your code or give ready-made solutions.

Thank you for following this guide ! 🥳

If you have any questions, don't hesitate to reach out to a member of the Hi!ckathon #6 team during the weekend.

And more importantly, **good luck** 😊

Hi!

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